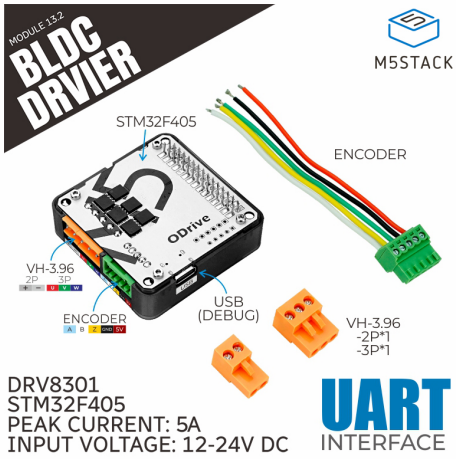
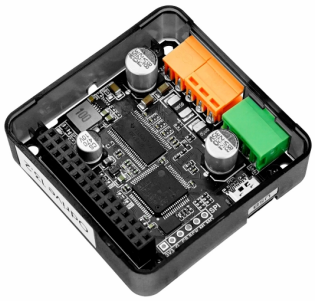
Module13.2 BLDC Drvier

SKU:M036

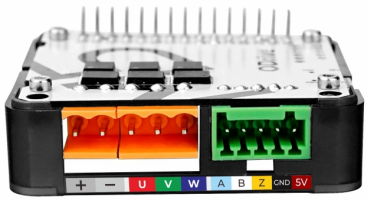














Description

Module13.2 BLDC Drvier is a high-performance servo motor driver module launched by M5Stack, based on the open-source motion control solution ODrive. It supports control of a single three-phase servo motor, with a peak drive current of up to 5A. While offering high-speed motor control capability, it also provides an encoder signal interface, which enables high-precision motion control and positioning. The module uses a UART communication interface and is compatible with the official ODrive configuration tool and protocol (via the host tool, various motor motion modes can be configured to make motor operation smoother and more stable).

Features

- Supports single three-phase servo motor drive
- Peak drive current 5A
- DC 12 ~ 24V power input interface (requires adapter output current up to 5A)
- Communication interface: UART
- Integrated encoder signal interface

Includes

- 1 x Module13.2 BLDC Drvier
- 1 x 3.96-3P Terminal
- 1 x 3.96-2P Terminal
- 1 x 2.54-5P Terminal (with cable)

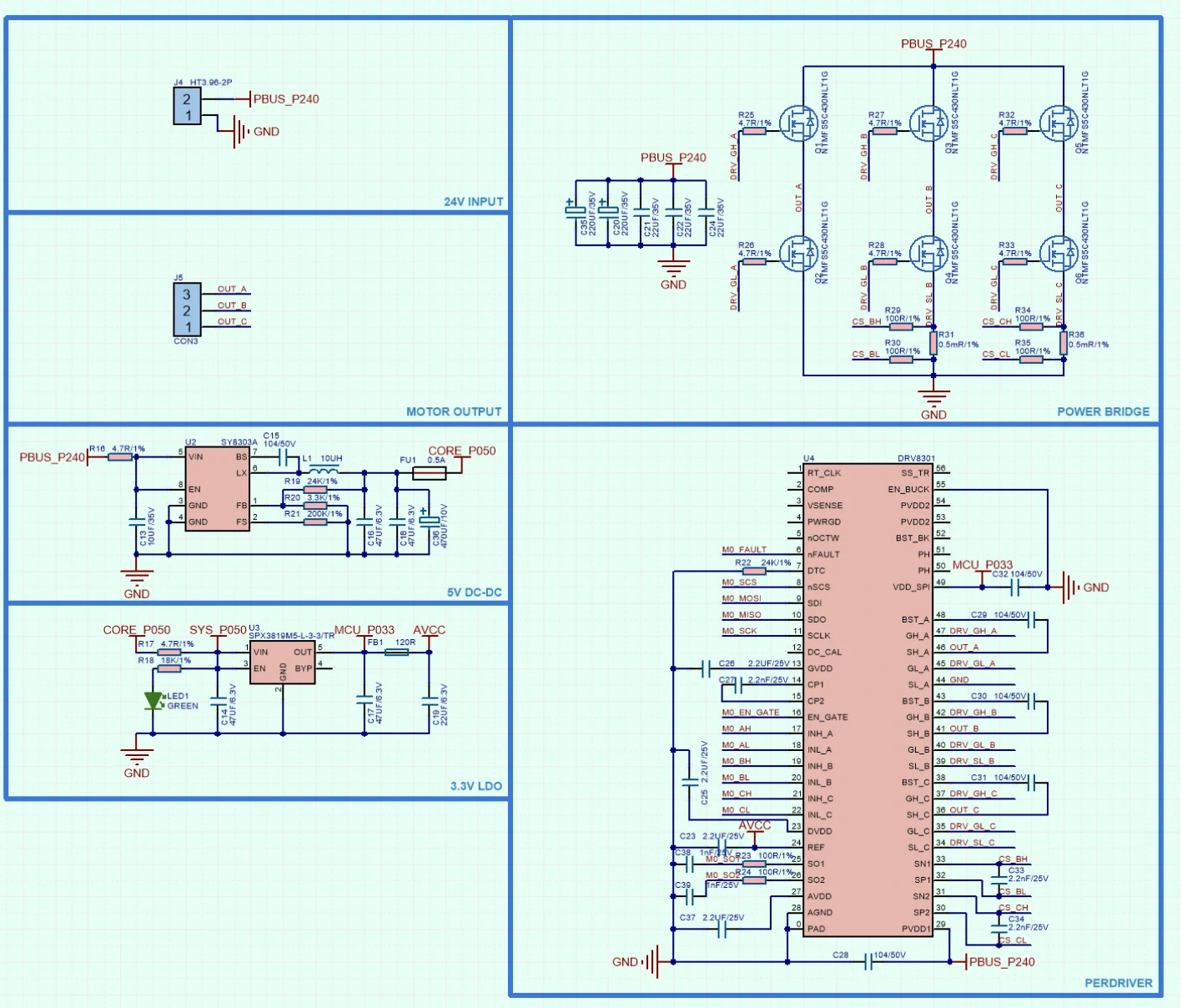
Applications

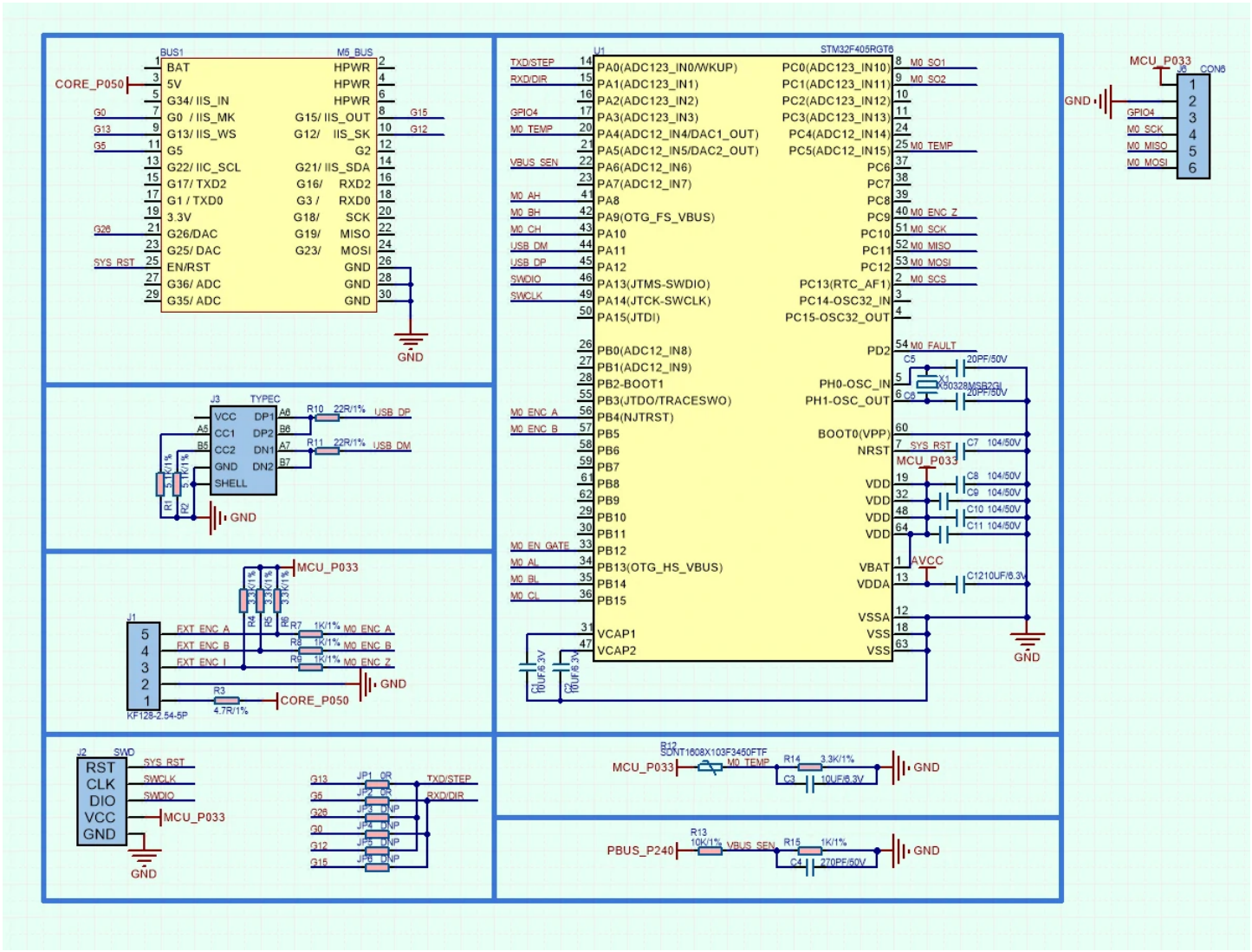
- High-precision motion control
- Servo motor drive

Specifications

Specification	Parameter
Maximum Drive Current	5A
Interface Type	3.96-2P (Power), 3.96-3P (Motor), 2.54-5P (Encoder)
ODrive Hardware Version	v3.5-24V
ODrive Firmware Version	0.5.1
Input Power Supply	DC 12 ~ 24V
Product Size	54.2 x 54.2 x 19.7mm
Product Weight	22.5g
Package Size	95.0 x 65.0 x 25.0mm
Gross Weight	42.3g

Schematics


24V INPUT
J4 HT3.96-2P
PBUS_P240
GND
MOTOR OUTPUT
J5
OUT_A
OUT_B
OUT_C
CON3
POWER BRIDGE
PBUS_P240
NTMFS5C430NLT1G
R25 4.7R/1%
R26 4.7R/1%
R27 4.7R/1%
R28 4.7R/1%
R32 4.7R/1%
R33 4.7R/1%
R29 100R/1%
R30 100R/1%
R31 0.5mR/1%
R34 100R/1%
R35 100R/1%
R36 0.5mR/1%
CS_BH
CS_BL
CS_CH
CS_CL
5V DC-DC
U2 SY8303A
R16 4.7R/1%
C15 104/50V
L1 10UH
FU1 0.5A
CORE_P050
R19 24K/1%
R20 3.3K/1%
R21 200K/1%
3.3V LDO
U3 SPX3819M5-L-3-3/TR
CORE_P050
SYS_P050
MCU_P033
AVCC
R17 4.7R/1%
R18 18K/1%
LED1 GREEN
FB1 120R
PERDRIVER
U4 DRV8301
RT_CLK
COMP
VSENSE
PWRGD
nOCTW
nFAULT
DTC
nSCS
SDI
SDO
SCLK
DC_CAL
GVDD
CP1
CP2
EN_GATE
INH_A
INL_A
INH_B
INL_B
INH_C
INL_C
DVDD
REF
SO1
SO2
AVDD
AGND
PAD
SS_TR
EN_BUCK
PVDD2
PVDD2
BST_BK
PH
PH
VDD_SPI
BST_A
GH_A
SH_A
GL_A
SL_A
BST_B
GH_B
SH_B
GL_B
SL_B
BST_C
GH_C
SH_C
GL_C
SL_C
SN1
SP1
SN2
SP2
PVDD1
M0_FAULT
M0_SCS
M0_MOSI
M0_MISO
M0_SCK
M0_EN_GATE
M0_AH
M0_AL
M0_BH
M0_BL
M0_CH
M0_CL
M0_SO1
M0_SO2
R22 24K/1%
R23 100R/1%
R24 100R/1%
C23 2.2UF/25V
C24 1nF/25V
C25 2.2UF/25V
C26 2.2UF/25V
C27 2.2nF/25V
C28 104/50V
C29 104/50V
C30 104/50V
C31 104/50V
C32 104/50V
C33 2.2nF/25V
C34 2.2nF/25V
C37 2.2UF/25V
C38 1nF/25V
C39 1nF/25V
MCU_P033
GND
PBUS_P240



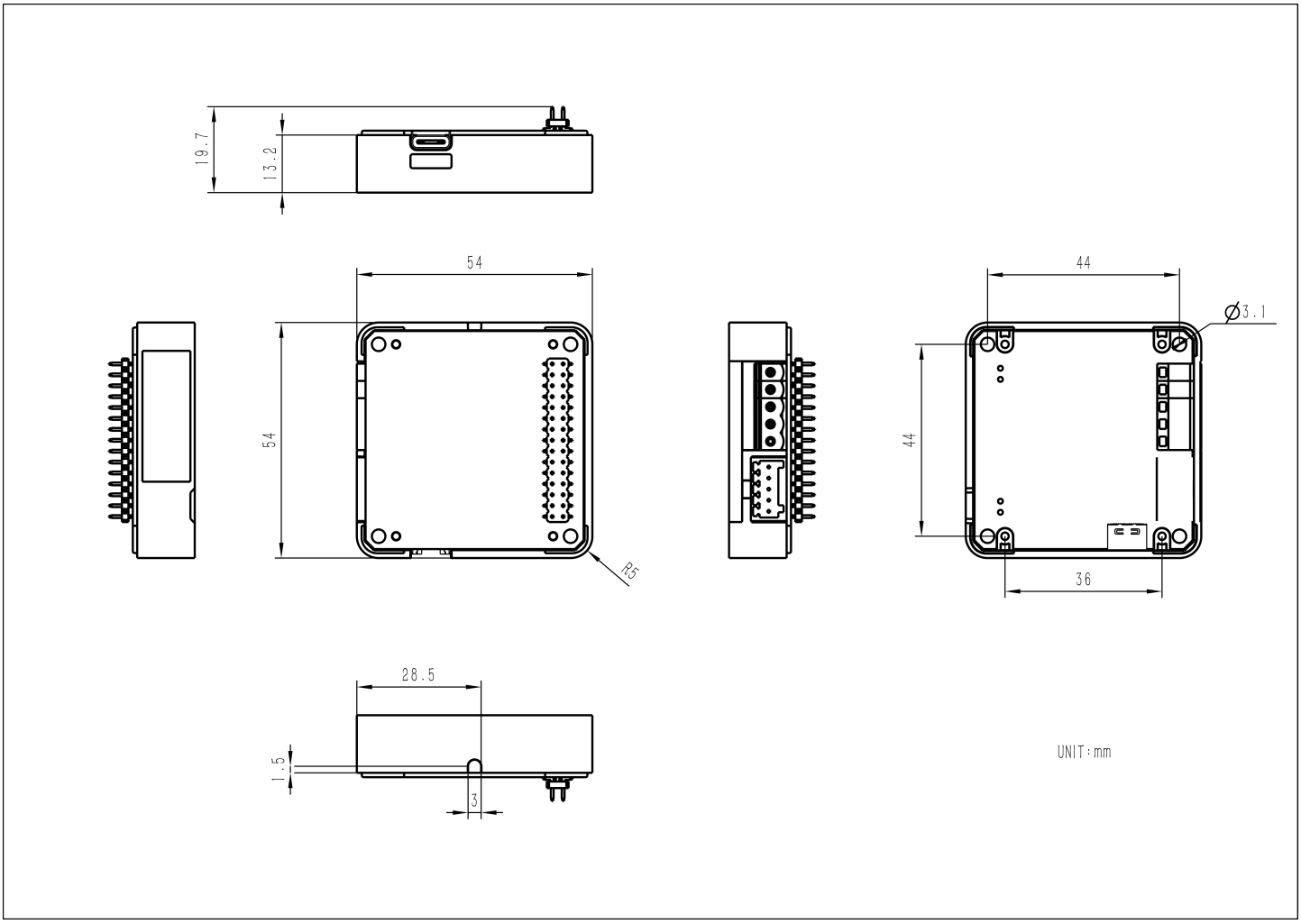
PinMap

M5-Bus

PIN	LEFT	RIGHT	PIN
GND	1	2	
GND	3	4	
GND	5	6	RST
	7	8	
	9	10	TXD / STEP
	11	12	
	13	14	
	15	16	
	17	18	
	19	20	RXD / DIR
TXD / STEP	21	22	TXD / STEP
RXD / DIR	23	24	RXD / DIR
	25	26	
	27	28	5V
	29	30	

Model Size

- Module13.2 BLDC Drvier Model Size PDF



Datasheets

- DRV8301 Datasheet
- Thermistor SDNT1608X103F3450FTF
- MOSFET NTMFS5C430NLT1G

Softwares

Arduino

Example Description

This example uses the ODrive module to control a servo motor for high-speed and precise rotation. Note: Before use, you need to configure relevant parameters using ODriveTool according to the connected motor model.

- Module13.2 BLDC Drvier Example with M5Core

UiFlow2

- Module13.2 BLDC Drvier UiFlow2 Docs

ODriveTool

> odrivetool is the configuration and debugging software paired with ODrive. It can be used to configure motor parameters. This tutorial demonstrates how to install and use odrivetool on the `Linux` platform.

- Use the following command to install odrivetool v0.5.1. Environment requirement: `python3`.

```
pip3 install odrive==0.5.1.post0
```

- Add `~/.local/bin` to the system environment variable. Execute the following command, and insert `export PATH=$PATH:~/.local/bin` to the end of the text.

```
vim ~/.bashrc
```

- Execute `odrivetool` in the command line to run the tool. Connect the ODrive module to the computer and wait for odrivetool to recognize it. After successful connection, enter `odrv0.vbus_voltage` to test and get the driver board voltage.

```
$odrivertool

ODrive control utility v0.5.1.post0
Website: https://odriverobotics.com/
Docs: https://docs.odriverobotics.com/
Forums: https://discourse.odriverobotics.com/
Discord: https://discord.gg/k3ZZ3mS
Github: https://github.com/madcowswe/ODrive/

Please connect your ODrive.
You can also type help() or quit().

Connected to ODrive 306A396A3235 as odrv0

In [1]: odrv0.vbus_voltage
```

- Common configuration commands.

```
// Configure motor current limit
odrv0.axis0.motor.config.current_lim [A].

// Configure motor speed limit
odrv0.axis0.controller.config.vel_limit

// Configure the resistance value for braking resistor
odrv0.config.brake_resistance

// Save configuration
odrv0.save_configuration()
```

- For more details, please check the official ODrive documentation

Video

ODrive_VIDEO.mp4